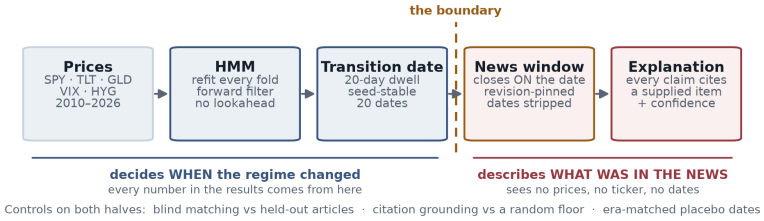
Regime detection that can say what happened — and a measurement of how much that is worth

An auditable explanation layer for any regime model. BSc Mathematics with Statistics, University of Bristol.

Finding: the explanations are specific and faithful to their sources; whether they are *diagnostic* of a regime change is unresolved, and this design could never have resolved it.



PROBLEM STATEMENT

1 · No explanation. A regime model tells you the market changed, not what happened — it only sees returns. A human fills the gap from memory, so every reading differs and none is written down.

2 · No verification. The obvious fix makes it worse. A language model answers in seconds, and it reads the same whether drawn from that month's news or invented. An unverifiable explanation beats no explanation only in the way that feeling informed beats being informed.

SOLUTION OVERVIEW

The HMM decides *when* the regime changed; the language model only describes *what was in the news* then. No statistic comes from the model. For each date it retrieves a 14-day news window and returns an explanation whose every claim cites a supplied article.

Nothing published after the date can enter the window — the code refuses to build one. Pages are pinned to the revision that existed then, because **38.2%** of a Wikipedia day-page (peak 71.2%) is written later.

USE OF AI

`claude-opus-5` via the Anthropic Messages API — one call per window, effort high, schema enforced server-side. Prompts are version-controlled with a dated iteration history, never inline strings; every call logs the model id, prompt hash and input hash, so any sentence traces to the exact prompt and news behind it. Built with Claude Code. MediaWiki API, yfinance, hmmlearn, scikit-learn, Streamlit.

VALIDATION

Can you trust the explanations?	What we found
Date hidden: can we still tell which two weeks it describes?	72% — guessing gives 33%
Claims really come from the source cited	96.4% — a random source gives 1.8%
Invented citations	none, in 474 claims
Confident on real events, vs ordinary days	**80% vs 62% — ignoring the news gives no gap**
Whole study re-run on a different AI	same scores

The detector underneath: stressed days move **2.18×** as much as calm days, in 18 of 20 folds independently ($p = 0.00040$).

IMPACT & VALUE

Who needs it. A risk team asked why a stress signal fired that week. A PM handing over a book they did not build. A researcher whose changepoint model produces dates nobody can interpret. All three rely on memory today.

What changes. The interpretation becomes a written record with dated sources, produced the same way every time, carrying a number for how far to trust it. **The reusable part is the controls, not the explanations** — point them at any regime model's dates, any asset, any method, and every explanation returns with the scorecard above attached.

Honest scope. Row four says these are not yet shown to tell real events from ordinary days — so this buys an auditable record of *why*, not a trading signal.

REFLECTIONS

The test that failed is the one I would keep — but it was never going to work. With only 20 real transitions it had a 17% chance of spotting a difference even if one was there; only more transitions would help, not more ordinary dates. **Next:** test only the days volatility *starts* — the days it *ends* carry no news driver, and that leaves ten cases, so it needs roughly three times the history. One gap stays open: the cut-off is midnight UTC; Wikipedia reports the US close that evening.